THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 2 (Suggested Solution)

1. (a) (**Bookwork**) See *Theorem 3.6*:

Theorem 3.6 (Cauchy Convergence Criterion for Series) The series $\sum_{n=1}^{\infty} a_n$ converges if and only if for any $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for any $n, m \in \mathbb{N}$, if $n > m \geq N$, then $|a_{m+1} + a_{m+2} + \cdots + a_n| < \epsilon$.

(b) (**Bookwork**) See *Theorem 3.9(a)*:

Given that $\sum_{n=1}^{\infty} b_n$ converges, it satisfies the Cauchy Convergence Criterion and so for any $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for any $n, m \in \mathbb{N}$, if $n > m \geq N$, then

$$b_{m+1} + b_{m+2} + \cdots + b_n = |b_{m+1} + b_{m+2} + \cdots + b_n| < \epsilon.$$

Therefore, if $n > m \geq N$, then

$$|a_{m+1} + a_{m+2} + \cdots + a_n| = a_{m+1} + a_{m+2} + \cdots + a_n \leq b_{m+1} + b_{m+2} + \cdots + b_n < \epsilon.$$

Thus the series $\sum_{n=1}^{\infty} a_n$ also satisfies the criterion and hence it converges.

(c) Note that for any α, $\beta \in \mathbb{R}$, we have $\alpha\beta \leq \frac{1}{2}(\alpha^2 + \beta^2)$. Now for any $n \in \mathbb{N}$, let $\alpha = \sqrt{a_n}$ and $\beta = \frac{1}{n}$. Then we have

$$\frac{\sqrt{a_n}}{n} \leq \frac{1}{2}\left(a_n + \frac{1}{n^2}\right) \quad \text{for any } n \in \mathbb{N}.$$

Given that $\displaystyle\sum_{n=1}^{\infty} a_n$ converges and we know that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, so $\sum_{n=1}^{\infty} \frac{1}{2}\left(a_n + \frac{1}{n^2}\right)$ converges. As $0 \leq \frac{\sqrt{a_n}}{n}$ for any $n \in \mathbb{N}$, by the Comparison Test, we conclude that $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n}$ converges.

2. (**Bookwork**) See *Assignment 5, Q.4*:

(a) We claim that $\lim_{x\to 0} \frac{1}{x} f\left(\frac{x}{n}\right) = \frac{1}{n} L$ for any $n \in \mathbb{N}$. Thus we want to show

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } \forall x \in \mathbb{R},\ \text{if } 0 < |x| < \delta,\ \text{then } \left|\frac{1}{x} f\left(\frac{x}{n}\right) - \frac{1}{n} L\right| < \epsilon. \qquad (*)$$

Given that $f(0) = 0$ and $f'(0) = L$, we have

$$L = \lim_{x\to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0} \frac{f(x)}{x}.$$

This means

$$\forall \epsilon' > 0,\ \exists \delta' > 0 \text{ such that } \forall x \in \mathbb{R},\ \text{if } 0 < |x| < \delta',\ \text{then } \left|\frac{f(x)}{x} - L\right| < \epsilon'. \qquad (**)$$

Now for any $\epsilon > 0$, consider $\epsilon' = n\epsilon > 0$. Then $(**)$ implies there exists $\delta' > 0$ such that for any $x \in \mathbb{R}$, if $0 < |x| < \delta'$, then

$$\left| \frac{f(x)}{x} - L \right| < n\epsilon.$$

Then we can choose $\delta = n\delta' > 0$. Thus for any $x \in \mathbb{R}$,

$$\begin{aligned} 0 < |x| < \delta \implies 0 < |x| < n\delta' \implies 0 < \left|\frac{x}{n}\right| < \delta' &\implies \left| \frac{f(\frac{x}{n})}{\frac{x}{n}} - L \right| < \epsilon' \quad \text{(by } (**)\text{)} \\ &\implies n \left| \frac{1}{x} f\left(\frac{x}{n}\right) - \frac{1}{n} L \right| < \epsilon' \\ &\implies \left| \frac{1}{x} f\left(\frac{x}{n}\right) - \frac{1}{n} L \right| < \epsilon. \end{aligned}$$

This proves $(*)$.

(b) It follows from the result in (a) that for any $n \in \mathbb{N}$,

$$\lim_{x \to 0} \frac{1}{x} \left(f(x) + f\left(\frac{x}{2}\right) + \cdots + f\left(\frac{x}{n}\right) \right) = \left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right) L.$$

3. (a) Given that $\lim_{x \to c} f'(x) = L$, we have for any $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in (a, b)$, if $x \in (c - \delta, c + \delta) \setminus \{c\}$, then

$$|f'(x) - L| < \epsilon.$$

Now for any $x \in (a, b)$ and $x \in (c - \delta, c + \delta) \setminus \{c\}$, we apply the Mean Value Theorem to f on $[x, c]$ or $[c, x]$. Then there exists $\xi \in (x, c)$ or (c, x) such that

$$f'(\xi) = \frac{f(x) - f(c)}{x - c}.$$

It follows that for any $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in (a, b)$, if $x \in (c - \delta, c + \delta) \setminus \{c\}$, then

$$\left| \frac{f(x) - f(c)}{x - c} - L \right| = |f'(\xi) - L| < \epsilon,$$

that is, f is differentiable at c with $f'(c) = L$.

(b) Consider the function $f : [-1, 1] \longrightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Then f is

- continuous on $[-1, 1]$;
- differentiable on $(-1, 1) \setminus \{0\}$;
- differentiable at 0 with $f'(0) = 0$.

Its derivative $f' : (-1, 1) \longrightarrow \mathbb{R}$ is given by

$$f'(x) = \begin{cases} 2x \sin \dfrac{1}{x} - \cos \dfrac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

However, $\lim_{x \to 0} f'(x)$ does not exist (because $\lim_{x \to 0} \cos \dfrac{1}{x}$ does not exist). This shows the converse of (a) is not true.

4. (a) <u>True</u>

- Since $\left|\dfrac{(-1)^{n+1}n}{n^2+1}\right| = \dfrac{n}{n^2+1} \geq \dfrac{n}{n^2+n} = \dfrac{1}{n+1}$ (≥ 0) for all $n \in \mathbb{N}$, and since the harmonic series diverges, the Comparison Test implies that the given series does not converge absolutely.
- Since $\dfrac{n}{n^2+1} \geq 0$ for all $n \in \mathbb{N}$, the sequence $\left\{\dfrac{n}{n^2+1}\right\}$ is decreasing, and $\lim\limits_{n\to\infty} \dfrac{n}{n^2+1} = 0$, thus, by alternating series test, the given series converges.

It follows that the given series converges conditionally.

(b) <u>True</u>

Assume to the contrary that f is continuous at some $c \in \mathbb{R} \setminus \{4\}$. Consider a sequence $\{x_n\}$ of rational numbers and a sequence $\{y_n\}$ of irrational numbers converging to this c (the existence of these sequences are well known). For any $n \in \mathbb{N}$, we have $f(x_n) = x_n^2$ and $f(y_n) = 8y_n - 16$. Hence

$$\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_n^2 = c^2 \quad \text{and} \quad \lim_{n\to\infty} f(y_n) = \lim_{n\to\infty} 8y_n - 16 = 8c - 16.$$

Since f is continuous at c, we must have $\lim\limits_{n\to\infty} f(x_n) = \lim\limits_{n\to\infty} f(y_n)$, that is, $c^2 = 8c - 16$, or equivalently, $c = 4$. This is a contradiction as $c \in \mathbb{R} \setminus \{4\}$.

(c) <u>False</u>

Assume to the contrary that there exists a continuous and bijective function $f : (0,1) \longrightarrow [0,1]$. Since f is surjective, there exist $x_0, x_1 \in (0,1)$ such that $f(x_0) = 0$ and $f(x_1) = 1$. WLOG, we may assume that $x_0 < x_1$.

Now consider a number between 0 and x_0, say $\frac{x_0}{2}$, and let $L = f(\frac{x_0}{2})$. Since f is injective, $L \neq 0\,(= f(x_0))$ and $L \neq 1\,(= f(x_1))$. Hence we have $0 < L < 1$. Then we can apply the Intermediate Value Theorem to f on $[x_0, x_1]$ (since f is continuous on $(0,1)$, and hence on $[x_0, x_1]$) and conclude that there exists $c \in (x_0, x_1)$ such that $f(c) = L$. This is a contradiction, as we have $f(\frac{x_0}{2}) = L = f(c)$ but $\frac{x_0}{2} \neq c$ (since $\frac{x_0}{2} < x_0 < c$).

This shows that there cannot be any continuous and bijective function $f : (0,1) \longrightarrow [0,1]$.

(d) <u>True</u>

Since $[a,b] \subseteq f([a,b])$, there exists $c, d \in [a,b]$ such that $f(c) = a$ and $f(d) = b$. If $c = a$ or $d = b$, then we are done. Thus we can assume that $c > a$ and $d < b$. Define $g : [a,b] \longrightarrow \mathbb{R}$ by $g(x) = f(x) - x$. Then g is continuous on $[a,b]$, and

$$g(c) = f(c) - c = a - c < 0 \quad \text{and} \quad g(d) = f(d) - d = b - d > 0.$$

Now we can apply the Intermediate Value Theorem to g on $[c,d]$ (or $[d,c]$) and conclude that there exists $x_0 \in [c,d]$ (or $[d,c]$) such that $g(x_0) = 0$, that is, $f(x_0) = x_0$. Hence the result.